

NUDGE — PROJECT DOCUMENTATION



Flutter Mobile Application
Course: Mobile Programming 2
Lecturer: Teng Rithrathanak

1. Project Overview

Nudge is a Flutter mobile application for location-based reminders. Users can create a reminder, choose a saved place, set a trigger type and radius, and manage their active nudges from one Home screen. The project focuses on a simple student-friendly design and a clear reminder flow.

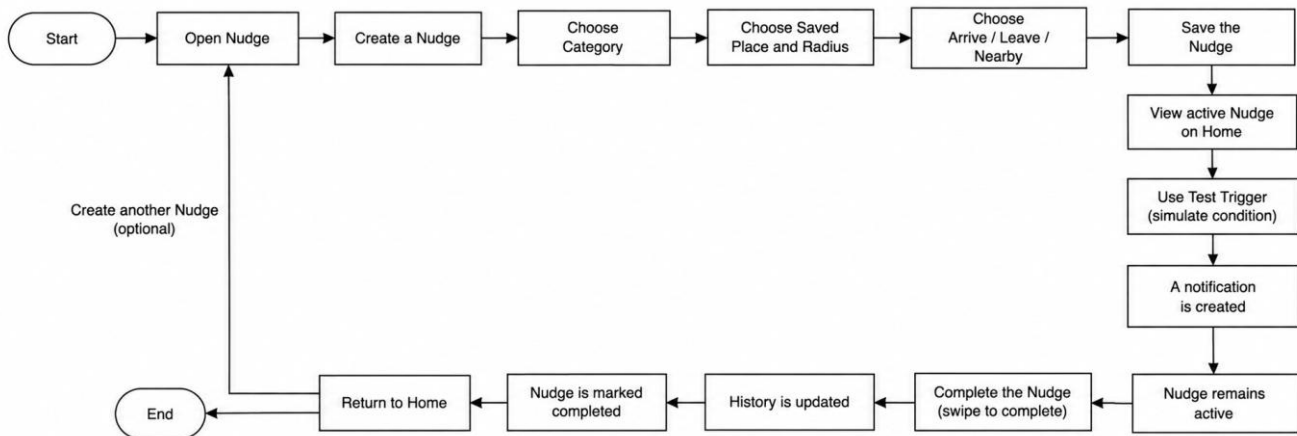
2. Project Objectives

- Make location reminders easy to create and manage.
- Allow users to save places and choose a trigger radius.
- Show notifications when an arrive, leave, or nearby trigger occurs.
- Keep the interface simple and support both light and dark mode.

3. Main Features

Create Nudge, Edit Nudge, Saved Places, Map Picker, Radius Settings, Categories, Arrive/Leave/Nearby triggers, Test Trigger simulation, History, Notifications, Appearance settings, and Home screen filtering/search.

4. User Flow



5. Technical Implementation

- Built with Flutter and Dart.
- Uses StatefulWidget and setState for screen state changes.
- Uses Navigator to connect the application's screens.
- Uses ListView.builder for scrollable lists and trigger content.
- Uses forms and basic validation for user input.
- Uses shared in-memory AppData for the current prototype.
- Uses NudgeTriggerService to keep trigger/completion logic in one place.
- Uses a shared theme system for light and dark mode.

6. Trigger Simulation

The current prototype does not run background GPS or real geofencing. To demonstrate the trigger behavior, the Test Trigger screen shows every active Nudge with an action based on its selected trigger. Arrival shows "Simulate Arrival," leaving shows "Simulate Leaving," and nearby shows "Simulate Nearby." After the action, a Notification record is created and the Nudge remains active. The user can then complete the Nudge manually from the Home screen.

7. Screen Summary

Screen	Purpose
Home	Shows active Nudges, search/filter controls, History, Notifications, and Test Trigger access.
New Nudge	Creates a Nudge by choosing a title, category, place, and trigger.
Edit Nudge	Updates an existing Nudge or deletes it.
Saved Places	Adds and edits saved places and their radius settings.
Map Picker	Selects a location and radius for a saved place.
Radius Settings	Manages available radius values and allows adding and deleting radius options.
History	Shows completed Nudge events with their time and related information.
Notifications	Shows notification records created when a trigger is simulated.
Test Triggers	Simulates Arrive, Leave, and Nearby trigger actions for active Nudges.
Settings	Provides appearance, categories, saved locations, radius, and About Nudge options.

8. Important App Behavior

A Nudge remains active on the Home screen after a trigger is simulated. The Test Trigger action creates a Notification but does not complete the Nudge. When the user swipes to complete the Nudge, a History entry is created and the Nudge is marked as completed and removed from the active list.

9. Testing

- Create and save Nudges with different categories, places, radius, and trigger types.
- Edit existing Nudges and verify that the selected place and trigger are updated correctly.
- Use Test Trigger for Arrival, Leaving, and Nearby actions.
- Check that a simulated trigger creates a Notification and the Nudge remains active.
- Complete the Nudge from Home and check that it appears in History and is removed from the active list.
- Check light and dark mode on the main screens.
- Run flutter analyze and verify that no analyzer issues are reported.

10. Current Limitations

This version is a prototype for the course. It does not use a backend, user account system, background GPS monitoring, or real geofencing. The Test Trigger screen is used to simulate trigger conditions during testing and presentation.

11. Team Responsibilities

- **Top Keitaro (Leader)**
Responsible for implementing the Home Page and Notification features, as well as contributing to overall application integration and development.
- **Chui Vanenteakbot**
Responsible for implementing the Settings feature and supporting the application's related functionality.
- **Sorn Kunwath**
Responsible for implementing the History feature and serving as the UI/UX Leader, overseeing the overall visual design, usability, and consistency of the application.
- **Lao David**
Responsible for implementing the Create Nudge feature, including the user flow for creating a new Nudge.

- **Keo Ratanak**

Responsible for assisting with UI/UX design and review under the direction of the UI/UX Leader. No major code implementation was assigned.

12. Conclusion

Nudge provides a simple way to create and manage location-based reminders while showing the main trigger and completion flow clearly. The prototype connects Nudge creation, saved places, trigger simulation, Notifications, completion, and History in one user flow.